

Reflection Journal

Name: Intentionally omitted

Date: 5/3/2021

Instructor: Samuel Chukwuemeka

Objective: The aim of the project was to...

- (1.) Present a dataset using a histogram.
- (2.) Describe the dataset.
- (3.) Determine the measures of central tendency of a dataset.
- (4.) Determine the measures of variation of a dataset.
- (5.) Determine the measures of location of a dataset.

Once the math of the dataset was completed, the aim was to:

- (1.) Display the dataset as an array in JavaScript.
- (2.) Develop a program that computes the Measures of Center, Measures of Spread, and Measures of Position of the dataset.

Company/Government: The Port Authority of New York and New Jersey

(<https://data.ny.gov/Transportation/Public-Parking-Counts-at-Port-Authority-of-NY-NJ-A/h87k-kqb6/data>)

Issues (if any):

The first issue I ran into was finding an easy way to calculate the mean of the six-digit numbers that composed the dataset I chose. Online I found a Long Addition Calculator: <https://www.calculatorsoup.com/calculators/math/long-addition-calculator.php>

The second issue I ran into was calculating the measures of variation: range, variance, and standard deviation. Fortunately, Professor Chukwuemeka's Descriptive Statistics Calculators were helpful.

The third issue I ran into was figuring out the lower and upper quartiles of my dataset. Online I found a Quartile Range Calculator:

<https://www.statisticshowto.com/calculators/interquartile-range-calculator/>

Description:

The project used data from the public parking counts at Kennedy International Airport. Specifically, the data (public parking counts) from month 10, 2015 to month 9, 2016.

Data Analysis:

Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016 (Raw Data)

Oct. 2015: 332,576	Apr. 2016: 326,042
Nov. 2015: 301,477	May 2016: 352,086
Dec. 2015: 338,196	June 2016: 375,360
Jan. 2016: 321,060	July 2016: 425,571
Feb. 2016: 269,176	Aug. 2016: 423,183
Mar. 2016: 313,331	Sep. 2016: 346,183

Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016 (Sorted Data in Ascending Order)

269,176	301,477	313,331	321,060
326,042	332,576	338,196	346,183
352,086	375,360	423,183	425,571

Draw the table with the class intervals.

Then, draw the frequency column and add the frequencies.

Check to make sure the sum of the frequencies is 12.

Public Parking Counts at Kennedy International Airport, from Oct. 2015 to Sep. 2016	Frequency (<i>f</i>)	$f * x$
269,176	1	269,176
301,477	1	301,477
313,331	1	313,331
321,060	1	321,060
326,042	1	326,042
332,576	1	332,576
338,196	1	338,196
346,183	1	346,183
352,086	1	352,086
375,360	1	375,360
423,183	1	423,183
425,571	1	425,571
	$\Sigma f = 12$	$\Sigma fx = 4,124,241$

Mean

$$(ii) \bar{x} = \frac{\Sigma fx}{\Sigma f}$$

$$\bar{x} = \frac{4,124,241}{12}$$

$$\bar{x} = 343,686.75$$

The average number of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, is approximately 343,687 counts.

Median

An even sample size means two middle numbers of the data sorted in ascending order.

They are the 6th and 7th data values, between the lowest and highest data.

6th data value (Oct. 2015) = 332,576

7th data value (Dec. 2015) = 338,196

$$\text{Median} = \frac{332,576 + 338,196}{2} = \frac{670,772}{2}$$

$$\text{Median} = 335,386$$

The median number of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, after the data has been sorted in ascending order, is 335,386 counts.

Mode

Data has no mode

Midrange

$$\text{Midrange} = \frac{\text{min} + \text{max}}{2}$$

$\text{min} = 269,176$ (Feb. 2016)

$\text{max} = 425,571$ (July 2016)

$$\text{Midrange} = \frac{269,176 + 425,571}{2} = \frac{694,747}{2}$$

$$\text{Midrange} = 347,373.5$$

February 2016 had the least number of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016: 269,176

July 2016 had the greatest number of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016: 425,571

The midrange number of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, is approximately 347,374 counts.

Range

$$\text{Range} = \text{max} - \text{min}$$

$\text{min} = 269,176$ (Feb. 2016)

$\text{max} = 425,571$ (July 2016)

$$\text{Range} = 425,571 - 269,176$$

Range = 156,395

February 2016 had the least number of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016: 269,176

July 2016 had the greatest number of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016: 425,571

The range of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, is 156,395 counts.

Variance

The variance is a quantity equal to the square of the standard deviation. The standard deviation is a way to measure the distance that dataset values are from the mean.

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\sum x = 4,124,241$$

$$n = 12$$

$$\bar{x} = 343,686.75$$

$x - \bar{x}$ for each value:

$$\text{Oct. 2015: } 332,576 - 343,686.75 = -11,110.75$$

$$\text{Nov. 2015: } 301,477 - 343,686.75 = -42,209.75$$

$$\text{Dec. 2015: } 338,196 - 343,686.75 = -5,490.75$$

$$\text{Jan. 2016: } 321,060 - 343,686.75 = -22,626.75$$

Feb. 2016: $269,176 - 343,686.75 = -74,510.75$

Mar. 2016: $313,331 - 343,686.75 = -30,355.75$

Apr. 2016: $326,042 - 343,686.75 = -17,644.75$

May 2016: $352,086 - 343,686.75 = 8,399.25$

June 2016: $375,360 - 343,686.75 = 31,673.25$

July 2016: $425,571 - 343,686.75 = 81,884.25$

Aug. 2016: $423,183 - 343,686.75 = 79,496.25$

Sep. 2016: $346,183 - 343,686.75 = 2,496.25$

$(x - \bar{x})^2$ for each value:

Oct. 2015: $-11,110.75^2 = 123,448,765.5625$

Nov. 2015: $-42,209.75^2 = 1,781,662,995.0625$

Dec. 2015: $-5,490.75^2 = 30,148,335.5625$

Jan. 2016: $-22,626.75^2 = 511,969,815.5625$

Feb. 2016: $-74,510.75^2 = 5,551,851,865.5625$

Mar. 2016: $-30,355.75^2 = 921,471,558.0625$

Apr. 2016: $-17,644.75^2 = 311,337,202.5625$

May 2016: $8,399.25^2 = 70,547,400.5625$

June 2016: $31,673.25^2 = 1,003,194,765.5625$

July 2016: $81,884.25^2 = 6,705,030,398.0625$

Aug. 2016: $79,496.25^2 = 6,319,653,764.0625$

Sep. 2016: $2,496.25^2 = 6,231,264.0625$

$\Sigma(x - \bar{x})^2$ is the sum of every value of $(x - \bar{x})^2$

$123,448,765.5625 + 1,781,662,995.0625 + 30,148,335.5625 + 511,969,815.5625$

$$\begin{aligned}
&+ 5,551,851,865.5625 + 921,471,558.0625 + 311,337,202.5625 + 70,547,400.5625 \\
&+ 1,003,194,765.5625 + 6,705,030,398.0625 + 6,319,653,764.0625 + \\
&6,231,264.0625 = 23,336,548,130.25
\end{aligned}$$

$$n - 1 = 12 - 1 = 11$$

$$s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1} = \frac{23,336,548,130.25}{11} = 2,121,504,375.4772727$$

$$s^2 = 2,121,504,375.4772727$$

The variance of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, is approximately 2,121,504,375 counts. This means there is a lot of spread between the values in the dataset.

(a lot of spread of the data values from the mean)

Standard Deviation

The standard deviation is a measure of the typical distance that values are from the mean. The bigger the standard deviation is, the more deviation there is between the values in the dataset.

$$\text{Standard Deviation} = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}$$

Look familiar? It is the same equation we used for the Variance.

Now all we need is to do is square s^2 to get s , the Standard Deviation.

$$s = \sqrt{2,121,504,375.4772727}$$

$$s = 46,059.79130952802$$

The standard deviation of Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, is approximately 46,060 counts. This means that there is a lot of spread between the dataset values.

(a lot of spread of the data values from the mean)

Measures of Location

The measures of location are found after the data has been sorted in ascending order. There are 12 numbers in the data set, so there are three numbers in each quartile (divided into four parts).

First Quartile: 269176 (Feb. 2016), 301477 (Nov. 2015), 313331 (Mar. 2016)

Second Quartile: 321060 (Jan. 2016), 326042 (Apr. 2016), 332576 (Oct. 2015)

Third Quartile: 338196 (Dec. 2015), 346183 (Sep. 2016), 352086 (May 2016)

Fourth Quartile: 375360 (June 2016), 423183 (Aug. 2016), 425571 (July 2016)

(For the interpretation of the Quartiles, include the condition: “when the dataset is divided into four equal parts”...)

The Lower Quartile (25th Percentile) is between 313,331 (Mar. 2016) and 321,060 (Jan. 2016)

$$\text{Lower Quartile} = \frac{313,331 + 321,060}{2} = \frac{634,391}{2}$$

Lower Quartile = 317,195.5

When the data for Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, is sorted in ascending order, approximately 317,196 counts are in the lower quartile, 25 percent.

We have already calculated the middle quartile: it is the same as the median.

When the data for Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, is sorted in ascending order, 335,386 counts are in the middle quartile, 50 percent.

The Upper Quartile (75th Percentile) is between 352,086 (May 2016) and 375,360 (June 2016)

$$\text{Upper Quartile} = \frac{352,086 + 375,360}{2} = \frac{727,446}{2}$$

Upper Quartile = 363,723

When the data for Public Parking Counts at Kennedy International Airport, from October 2015 to September 2016, is sorted in ascending order, 335,386 counts are in the middle quartile, 75 percent.

Five-Number Summary of Data

Minimum (min) = 269,176 (Feb. 2016)

Lower Quartile = 317,195.5

Middle Quartile (Median) = 335,386

Upper Quartile = 363,723

Maximum (max) = 425,571 (July 2016)

Benefit of the Project (if any): The project conveniently shows mean, median, mode, midrange, range, variance, standard deviation, and the quartiles of the dataset. This could help Kennedy International Airport figure out if they need more space for parking based on the data. It also gives an idea of what months people are more or less likely to travel during.

Limitation (if any): The project does not show mean deviation of the dataset.

Recommendation (if any):

Conclusion: The dataset is meant to represent a year of Public Parking Counts at Kennedy International Airport. Being an international airport, traffic in and out of it is heavy. The numbers for the dataset are huge, but varied. The least number of public parking counts is during the month of February, while the greatest number of public parking counts is during the month of July.

References:

Chukwuemeka, Samuel. "Descriptive Statistics Calculators." *Descriptive Statistics Calculators*, Samdom4Peace Designs, 2017, <https://descriptive-statistics.appspot.com/descriptiveStatistics.html>.

Furey, Edward. "Long Addition Calculator." *CalculatorSoup*, n.d., <https://www.calculatorsoup.com>

On the right is an example of how the long addition calculator works.

Glen, Stephanie. "Interquartile Range Calculator" *StatisticsHowTo.com*, 2021, <https://www.statisticshowto.com/calculators/interquartile-range-calculator/>.

The Port Authority of New York and New Jersey. "Public Parking Counts at Port Authority of NY NJ Airports: Beginning 2002." *State of New York*, 2017, data.ny.gov/Transportation/Public-

Long Addition Calculator

Enter numbers to add:

269176
301477
313331
321060
326042
332576
338196
346183
352086
375360
423183
425571

}

+

= ?

☐ Show long addition steps

Clear
Calculate

Answer:

carry	4	4	5	3	8	4	
		2	6	9	1	7	6
		3	0	1	4	7	7
		3	1	3	3	3	1
		3	2	1	0	6	0
		3	2	6	0	4	2
		3	3	2	5	7	6
		3	3	8	1	9	6
		3	4	6	1	8	3
		3	5	2	0	8	6
		3	7	5	3	6	0
		4	2	3	1	8	3
+		4	2	5	5	7	1
		4	1	2	4	2	4

Parking-Counts-at-Port-Authority-of-
NY-NJ-A/h87k-kqb6/data.